

A new look at reactions of 2-butoxy and 1-butoxy radicals in the presence of oxygen

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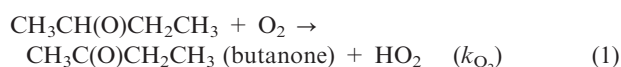
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Under atmospheric conditions, experiments show that 2-butoxy radicals in the presence of oxygen yield acetaldehyde and butanone such that the concentration ratio [acetaldehyde]/[O₂]/[butanone] shows a linear dependence on oxygen concentration [O₂]. [Zabel *et al.*, *Phys. Chem. Chem. Phys.*, 2002, **4**, 2579; Cox *et al.* *Phys. Chem. Chem. Phys.*, 2005, **7**, 3702.] A similar oxygen dependence was found by Cox *et al.* in the reactions of 1-butoxy radicals. These experiments imply a unimolecular rate constant that apparently depends on oxygen pressure. Previously this has been explained by postulating the presence of a small amount of excited radicals assumed to undergo 'prompt' dissociation. We propose an alternative interpretation based on the solution of the time-dependent master equation that does not require the putative presence of excited radicals. We do this by allowing the time to run until the so-called long-time steady-state, and then show that the oxygen dependence arises quite naturally if the transport matrix takes into account that in addition to the principal reaction channel (dissociation into acetaldehyde in the case of 2-butoxy and isomerization in the case of 1-butoxy), there is a small "leak" of alkoxy radicals due to the reaction with oxygen, and provided the calculation takes into account that in the cited experiments there is a continuous incoming flux of butoxy radicals.

Introduction

Under atmospheric conditions, the alkoxy radicals may either react with O₂ (if an α -H atom is present), or undergo unimolecular dissociation *via* a C–C bond scission. If the alkoxy radical has a side-chain with four or more than four carbon atoms, which is the case of the 1-butoxy radical, isomerisation occurs instead. In the case of 2-butoxy radicals with C₃ side-chain, a reaction with O₂ competes with unimolecular dissociation:



There are two relative-rate studies^{1–2} of the dissociation of 2-butoxy radicals using the reaction with O₂ as a reference, done in a static¹ or slow-flow² photoreactor with detection of products by IR absorption¹ or gas chromatography with flame ionisation detection (GC-FID).² The two methods were used to generate alkoxy radicals RO, both of which can provide excess energy to the RO.

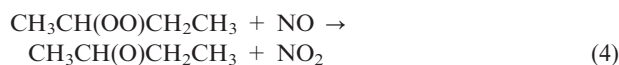
(i) Alkoxy radicals RO may be directly produced by near-UV photolysis of the alkyl nitrite (RONO) precursor. This method was used by Zabel *et al.*¹ who reported the results of

two experiments performed at 298 K and atmospheric pressure of air (800 mbar N₂, 200 mbar O₂), one by (continuous) photolysis at $\lambda = 254$ nm, and the other with a broad-spectrum source at $\lambda = 320$ –400 nm. In a later more extensive series of experiments, Cox *et al.*² have irradiated the nitrite at $\lambda = 360 \pm 10$ nm, at atmospheric pressure and at several temperatures in the range of 256–310 K, with variable oxygen concentration in the range $(0.05\text{--}2) \times 10^{19}$ molecule cm^{–3}. In both studies the experimental results were given by R_{exp} , the ratio of the yields of acetaldehyde and butanone multiplied by the oxygen concentration [O₂]

$$R_{\text{exp}} = ([\text{CH}_3\text{CHO}][\text{O}_2])/2[\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3] \quad (3)$$

plotted in Fig. 1 as R_{exp} vs. [O₂] (symbols).

(ii) Alkoxy radicals may also be generated indirectly by the exothermic reaction of alkyl peroxy radical with NO:



Using this method, Zabel *et al.*¹ reported two sets of experiments. The first set was done at atmospheric pressure, in the temperature range 280–313 K with oxygen concentration varied in the range $(0.2\text{--}3) \times 10^{19}$ molecule cm^{–3}. These results are shown in Fig. 2 likewise in the form R_{exp} vs. O₂ concentration (empty symbols). The second set of experiments was done at 298 K at constant partial pressure of O₂ ($P_{\text{O}_2} = 100$ mbar) and various total pressures ($P_{\text{tot}} = 100$ –1000 mbar), as well as in pure oxygen ($P_{\text{O}_2} = P_{\text{tot}}$); the results are shown in Fig. 3 (symbols). Similar experiments were carried out by Cox *et al.*² at atmospheric pressure, at 298 and 283 K, in the same range

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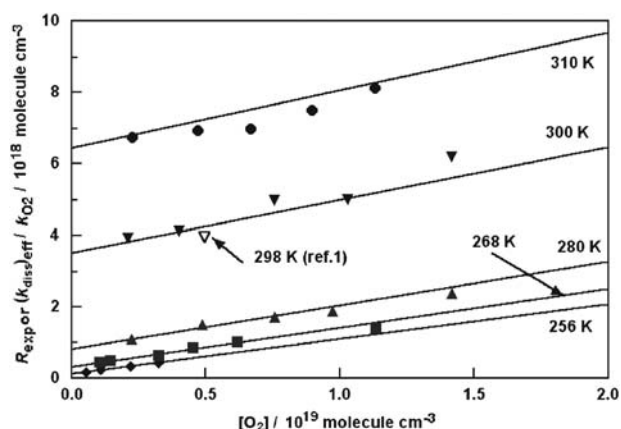


Fig. 1 Results of the dissociation of a 2-butoxy radical produced by photolysis of 2-butylinitrite at 360 nm, as a function of the partial pressure of oxygen at different temperatures. The symbols are experimental data R_{exp} from Fig. 2 in ref. 2; one symbol at 298 K is from ref. 1. The lines are present calculated results for $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$.

of oxygen concentrations; these experimental results are also shown in Fig. 2 (filled symbols).

From steady-state kinetic analysis of the reaction sequence (1)–(2), assuming that subsequent reactions of the C_2H_5 radicals lead to the formation of a second acetaldehyde molecule, and neglecting the possible formation of excited alkoxy radicals (*vide infra*), the following relationship is obtained²

$$([\text{CH}_3\text{CHO}][\text{O}_2])/2[\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3] = k_{\text{diss}}/k_{\text{O}_2} \quad (5)$$

According to eqns (3) and (5), the experiment should yield a constant R_{exp} at a given temperature. However, all results in Fig. 1 and 2 show that at every temperature R_{exp} increases linearly with increasing concentration of O_2 .

About ten years ago, in a study on atmospheric fate of HFC-134a, Wallington *et al.*³ studied the two competing reaction pathways for RO alkoxy radicals: decomposition or the reaction with O_2 . They found that decomposition was

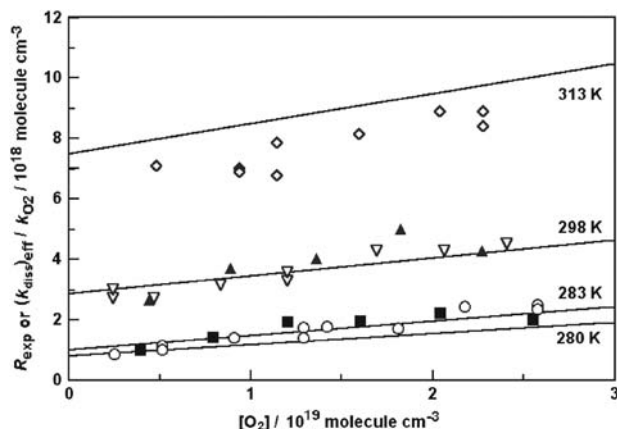


Fig. 2 Results for the dissociation of a chemically activated 2-butoxy radical produced in the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$, as a function of the partial pressure of oxygen at different temperatures. The symbols are experimental data R_{exp} : empty symbols, from Fig. 3 in ref. 1; filled symbols from Fig. 4 in ref. 2. The lines are present calculated results for $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$.

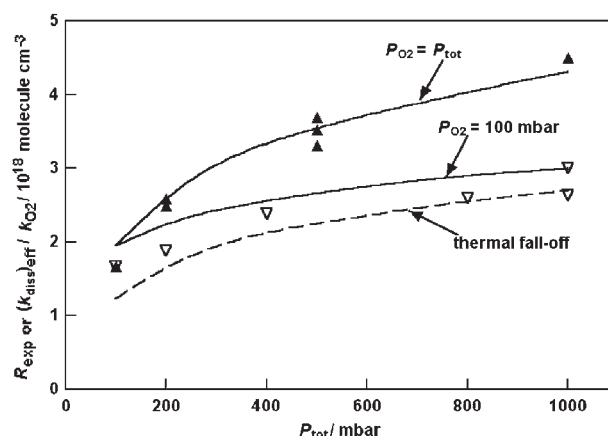
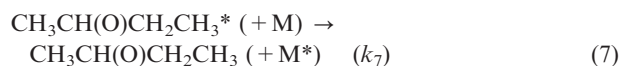


Fig. 3 R_{exp} vs. total pressure at 298 K for the 2-butoxy radical produced in the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. The symbols are experimental data R_{exp} from ref. 1. The curves represent the present calculations of $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$. The dotted curve is the calculated thermal fall-off of $k_{\text{diss}}/k_{\text{O}_2}$.

much more important when the RO radicals were produced *via* the exothermic reaction $\text{RO}_2 + \text{NO}$ than when they were produced *via* the thermoneutral self-reaction $\text{RO}_2 + \text{RO}_2$. Taking the clue, Peeters *et al.*⁴ and later we⁵ have dealt with the interpretation of these experimental results on the premise that there were two forms of alkoxy radicals: ‘hot alkoxy’ that dissociate by ‘prompt dissociation’ almost instantaneously relative to the reaction with O_2 , and stabilised radicals which thermally dissociate in competition with the reaction with O_2 . The slight linear dependence of the measured dissociation rate constant with the O_2 partial pressure was then explained by the formation of excited RO^* alkoxy radicals.⁶

Thus, it is now acknowledged^{1,2,5} that the dependence of R_{exp} on oxygen concentration can be interpreted as the consequence of the production of a small amount (6–10%) of vibrationally excited alkoxy radicals, denoted below by $\text{CH}_3\text{CH}(\text{O})\text{CH}_2\text{CH}_3^*$. In the cited experimental studies^{1,2} the proposed mechanism involves the formation of energy-rich alkoxy radicals that opens the possibility of a ‘prompt’ unimolecular reaction which competes with collisional relaxation and with which the reaction with O_2 cannot compete. For the 2-butoxy radicals two important reactions must be added:



A steady-state solution² of the somewhat complicated full mechanism then leads to

$$[\text{CH}_3\text{CHO}][\text{O}_2]/2[\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3] = (k_{\text{diss}}/k_{\text{O}_2})_{\text{eff}} \quad (8)$$

where

$$\left(\frac{k_{\text{diss}}}{k_{\text{O}_2}}\right)_{\text{eff}} = \frac{k_{\text{diss}}}{k_{\text{O}_2}} \left(1 + \frac{k_6}{k_7}\right) + \frac{k_6}{k_7} [\text{O}_2] \quad (9)$$

which now predicts linear dependence of $(k_{\text{diss}}/k_{\text{O}_2})_{\text{eff}}$ on oxygen concentration, with non-zero intercept, as is effectively observed for R_{exp} (*cf.* Fig. 1 and 2).

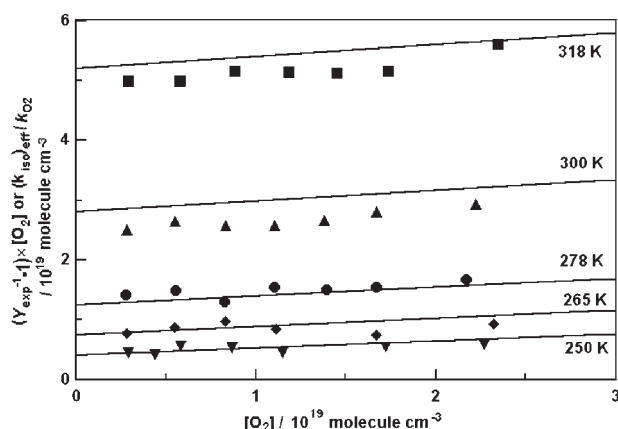
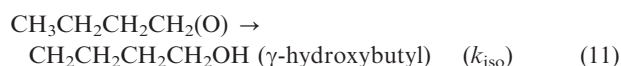
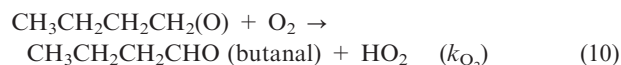


Fig. 4 Results of the isomerisation of the 1-butoxy radical produced by photolysis of 1-butylnitrite at 360 nm, as a function of the partial pressure of oxygen at different temperatures. The symbols are experimental data: $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2]$ (see text) from Fig. 3 in ref. 2. The lines are present calculated results for $(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2}$.

Similar experimental work was done by Cox *et al.*² on the unimolecular isomerisation of the 1-butoxy radical using the same procedures to produce the radical. In this case the two competing reactions are:



As the only product identified and quantified is butanal, the product of the reaction of 1-butoxy with O_2 , the analytical procedure gives Y_{exp} , the yield of butanal, instead of a ratio similar to R_{exp} . A steady-state treatment² leads to a slightly more complicated relation between $(k_{\text{iso}}/k_{\text{O}_2})_{\text{eff}}$ and Y_{exp} : $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2] = (k_{\text{iso}}/k_{\text{O}_2})_{\text{eff}}$ (see expression (ii) in ref. 2 for production of alkoxy by photolysis of nitrite). In Fig. 4, the corresponding experimental results are plotted in the form $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2]$ vs. $[\text{O}_2]$. The reason advanced at the time for the dependence on oxygen concentration was likewise attributed to a fraction of 1-butoxy radicals that react ‘promptly’. The experiments cannot confirm this fact for 1-butoxy radicals produced by the reaction of alkyl peroxy with NO.

A theoretical modelling of these phenomena, using an RRKM-Master Equation (RRKM-ME) approach was previously proposed by us^{5,6} to calculate the percentage of ‘prompt’ reaction. We now propose an alternative interpretation based on the solution of the time dependent Master Equation.

Theory

The alkoxy radicals disappear by two simultaneous pathways, one of which is an unimolecular reaction, dissociation (2-butoxy radical) or isomerisation (1-butoxy radical). This pathway is treated as standard unimolecular reactions with a microcanonical rate constant $k_{\text{uni}}(E)$ having a reaction threshold at E_0 . The conventional RRKM approach was used to calculate $k_{\text{uni}}(E)$ for the 2-butoxy radicals dissociation. In the

case of isomerisation of the 1-butoxy radical the tunneling effect was taken into account using semi-classical WKB approximation to calculate the tunneling probability, and the corresponding $k_{\text{uni}}(E)$ was calculated as described in ref. 7. The other pathway is the bimolecular reaction between an alkoxy radical and oxygen with a bimolecular rate constant k_{O_2} for which Atkinson⁸ gives $k_{\text{O}_2} = 1.5 \times 10^{-14} \exp(-200/T)$ or $k_{\text{O}_2} = 6 \times 10^{-14} \exp(-550/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ for the 2-butoxy and 1-butoxy radicals, respectively (eqns (1) and (10)). Both are thus essentially independent of temperature, so that to a first approximation each $k_{\text{O}_2}[\text{O}_2]$ may be considered to represent an energy-independent first-order rate constant.

Hence the total microcanonical rate constant for the reaction of an alkoxy radical of specified energy E is $k_{\text{uni}}(E)$, plus the energy-independent term $k_{\text{O}_2}[\text{O}_2]$.

The basic equation for time-dependent chemical activation is⁹

$$\frac{\partial n(E, t)}{\partial t} = F(E, t) + \omega \int_0^\infty p(E, E') n(E', t) dE' - [\omega + k(E)] n(E, t) \quad (12)$$

where $n(E, t)$ is the fractional population of reactant molecules of energy E at time t , $\omega = Z_{\text{LJ}}[M]$ is the collision frequency where Z_{LJ} is the Lennard-Jones collision number and $[M]$ the total gas concentration, $p(E, E')$ is the probability, per collision and per unit energy, for transfer from initial energy E' to final energy E . A simple exponential model is used for $p(E, E')$ where the probability of energy loss $\Delta E = |E - E'|$ in a collision is proportional to $\exp(-\Delta E/\langle \Delta E \rangle_{\text{down}})$ with $\langle \Delta E \rangle_{\text{down}}$ a temperature independent factor.

If $k_{\text{uni}}(E)$ is the microcanonical constant for unimolecular dissociation of 2-butoxy radicals having energy $E \geq E_0$, then in the present context, for energies at and above E_0 , $k(E) = k_{\text{uni}}(E) + k_{\text{O}_2}[\text{O}_2]$, and $k(E) = k_{\text{O}_2}[\text{O}_2]$ below E_0 . In the case of the isomerisation of 1-butoxy, due to the tunneling effect, the unimolecular reaction can take place, in principle, for all energies below E_0 although with vanishing $k_{\text{uni}}(E)$ for E much below E_0 .

$F(E, t) = \mathcal{R}f(E, t)$ is the fractional incoming flux of the reactant at time t , $\mathcal{R}$ is the overall rate of alkoxy formation and $f(E, t)$ the normalised initial population distribution resulting from either photoactivation or chemical activation (*vide infra*). Under the experimental conditions dealt with here, the incoming flux is assumed time-independent.

After symmetrization, the basic equation can be written in matrix notation in the form

$$\frac{d|N(t)\rangle}{dt} = |F(t)\rangle + \omega \mathbf{J}_s |N(t)\rangle \quad (13)$$

where $|N(t)\rangle$ is a column vector with elements of $n(E, t)/B(E)^{1/2}$ with $B(E)$ the normalised Boltzmann distribution appropriate to the temperature of the process; similarly the column vector $|F(t)\rangle$ has elements $F(E, t)/B(E)^{1/2}$. The square symmetric matrix $\mathbf{J}_s$ represents symmetrized elements of $\int p(E, E') n(E', t) dE' - \{1 + (k_{\text{uni}}(E) + k_{\text{O}_2}[\text{O}_2])/\omega\}$.

The solution of eqn (13), assuming initially $N(0) = 0$, as is usual in chemical activation, is:

$$|N(t)\rangle = \int_0^t dz \exp[\omega \mathbf{J}_s(t-z)] |\mathbf{F}(z)\rangle \quad (14)$$

Following Schranz and Nordholm¹⁰ the exponential is expanded and transformed using expansion in terms of the eigenvector matrix $\mathbf{F}$ and dimensionless eigenvalues μ_k of the matrix $\mathbf{J}_s$. The final result for the dimensionless population vector $|N(t)\rangle$, with the simplification, appropriate to the experimental conditions dealt with here, namely that the incoming flux $\mathbf{F}$ is time-independent, is

$$|N(t)\rangle = \sum_j \frac{a_j}{\omega |\mu_j|} [1 - \exp(\omega \mu_j t)] |f_j\rangle \quad (15)$$

where $a_j = \langle f_j | \mathbf{F} \rangle$.

From the time-dependent energy distribution of the alkoxy radical $N(E, t)$ obtained from eqn (15), the time-dependent fraction of input alkoxy flux that disappears $D(t)$, either by a unimolecular reaction or by reaction with O_2 , is obtained

$$D(t) = \frac{1}{\omega} \sum_{E_0}^{\infty} k(E) B(E)^{1/2} N(E, t) \quad (16)$$

$D(t)$ was calculated under conditions corresponding to the experiments considered here (total pressure, temperature, oxygen partial pressure and level of excitation). Three different time regimes can be distinguished, cf. Fig. 5: (i) incubation period, when the reactant population is building up, which occurs extremely quickly on a time-scale on the order of a single collision (10^{-12} – 10^{-9} s); (ii) intermediate steady-state between 10^{-9} and 10^{-7} s characterised by a time-independent reactive fraction of the input flux; (iii) long-time steady state corresponding to times in excess of 10^{-4} s, with a stationary energy distribution population of the reactants (the total input flux equals the total out flux) and a zero stabilisation flux. In the experiments of Zabel *et al.*¹ and Cox *et al.*,² a detection of the products, and therefore the determination of the ratio R_{exp} (or Y_{exp}), was carried out for times of about 10 s and 0.5 min,

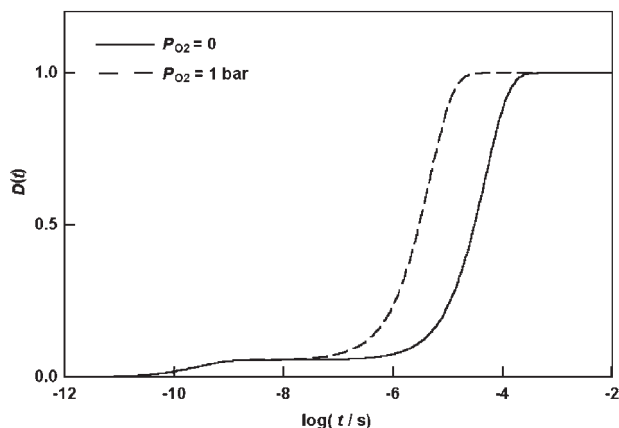


Fig. 5 Time-dependent fraction decomposed ($D(t)$, eqn (16)) of the incoming flux of the 2-butoxy radical produced from reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. $T = 298$ K; $P_{\text{tot}} = 1$ bar. Solid line $P_{\text{O}_2} = 0$; dashed line, $P_{\text{O}_2} = 1$ bar.

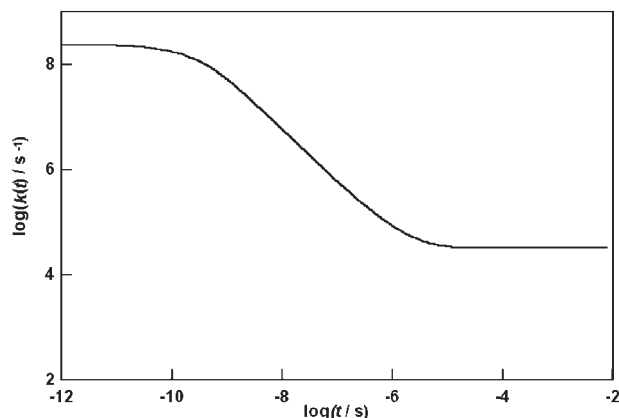


Fig. 6 Time-dependent dissociation rate constant ($k(t)$, eqn (17)) for 2-butoxy radicals produced from reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. $T = 298$ K; $P_{\text{tot}} = P_{\text{O}_2} = 1$ bar.

respectively, so that the intermediate steady state is of no direct interest in the present context. The actual experimental conditions correspond instead to the final long-time steady-state, where $D(t \rightarrow \infty) = 1$.

Next, we define a time-dependent unimolecular rate constant $k(t)$ (s^{-1}) by

$$k(t) = \frac{\sum_{E_0}^{\infty} k_{\text{uni}}(E) N(E, t)}{\sum_{E_0}^{\infty} N(E, t)} \quad (17)$$

The corresponding plot of $k(t)$ vs. t , calculated under the same experimental conditions as for $D(t)$ in Fig. 5, is shown in Fig. 6. At long times, $k(t)$ approaches the limit $(k_{\text{diss}})_{\text{eff}}$ (dissociation of 2-butoxy) or $(k_{\text{iso}})_{\text{eff}}$ (isomerisation of 1-butoxy) which we identify with the effective rate constant experimentally observed. After division by k_{O_2} we calculate $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ or $(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2}$, using the k_{O_2} values from Atkinson;⁸ thus

$$k(t \rightarrow \infty)/k_{\text{O}_2} \equiv (k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2} \text{ for dissociation of 2-butoxy} \quad (18)$$

$$\text{or } k(t \rightarrow \infty)/k_{\text{O}_2} \equiv (k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2} \text{ for isomerisation of 1-butoxy} \quad (18')$$

which we take to represent the calculated counterpart of the experimental R_{exp} or $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2]$ for 2-butoxy dissociation and 1-butoxy isomerisation, respectively.

Energy distribution of incoming flux of alkoxy radical

The energy distribution of the incoming flux of alkoxy radicals $f(E)$ depends on the way the radicals are produced. As mentioned, two ways were used in the experiments:^{1,2} (i) chemically activated alkoxy radical obtained from the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$, (ii) photoactivated alkoxy produced from the photolysis of the corresponding nitrite.

Calculations of the distribution of RO radicals produced by chemical activation in reaction (4) were detailed in ref. 5. Fig. 7 shows this initial ($t = 0$) internal energy distribution for the 2-butoxy radical at 298 K and total pressure 1 bar (dashed curve). It is a fairly wide distribution with a maximum at 3800 cm^{-1} .

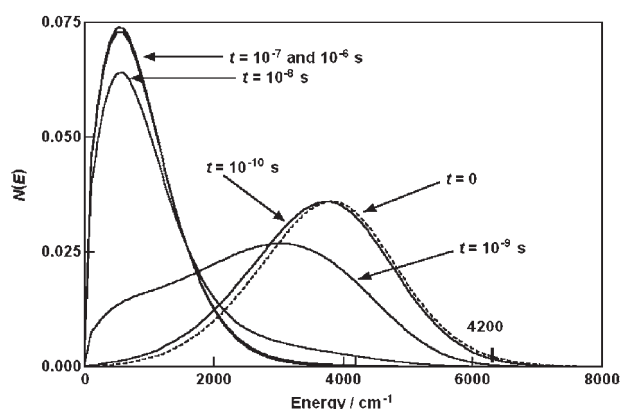


Fig. 7 Calculated time evolution of population distribution $N(E)$, eqn (15) of a chemically activated 2-butoxy radical produced in the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. $T = 298 \text{ K}$; $P_{\text{tot}} = 1 \text{ bar}$. Reaction threshold is at 4200 cm^{-1} .

The internal energy of RO radicals produced by photolysis of RONO may be deduced from the work of Lahmani *et al.*¹¹ on the photofragmentation of CH_3ONO at 355 nm. They have shown that about 38% of the available energy shows up in the CH_3O radical. This is much less than could be accounted for on a purely statistical basis, which can be reasonably expected to hold also for the fragmentation producing the larger RO radical of the size of butoxy. With internal energy of the parent butylnitrite molecule $E_{\text{int}}(\text{RONO}) \approx 1000 \text{ cm}^{-1}$ and an estimated dissociation energy $D_0(\text{RONO}) = 14\,500 \pm 400 \text{ cm}^{-1}$ ¹² the available energy for excitation at 370 nm ($27\,000 \text{ cm}^{-1}$), may be deduced to be $E_{\text{av}} = 27\,000 + 1000 - 14\,500 \approx 13\,500 \text{ cm}^{-1}$. In the absence of any other information, we use the data given by Lahmani *et al.*¹¹ for CH_3ONO , so that the order of magnitude of energy carried off initially by the butoxy RO radicals we estimate to be 38% of $13\,500 \text{ cm}^{-1}$ which works out to $\approx 5000 \text{ cm}^{-1}$, assumed to be randomized among the internal degrees of freedom of the excited species. We take this initial distribution to be a delta function at $E = 5000 \text{ cm}^{-1}$ as shown in Fig. 8 (dashed line), admittedly a simplification. As good agreement was obtained between

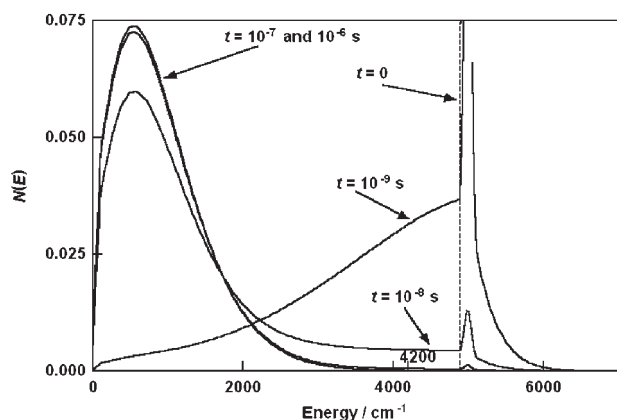


Fig. 8 Calculated time evolution of the population distribution $N(E)$ of the 2-butoxy radical produced in the photolysis of 2-butylnitrite at 360 nm. $T = 298 \text{ K}$; $P_{\text{tot}} = 1 \text{ bar}$. The initial distribution is a delta function at 5000 cm^{-1} . Reaction threshold is at 4200 cm^{-1} .

calculation and experiment (*vide infra*), this distribution was retained.

Molecular properties

The structural parameters of 2-butoxy radicals and of the corresponding transition state for dissociation were presented in ref. 13. With a threshold energy $E_0 = 50.2 \text{ kJ mol}^{-1} \equiv 4200 \text{ cm}^{-1}$, the RRKM calculated value of the thermal dissociation rate constant at 298 K and 1 bar is $k_{\text{diss}} = 2.07 \times 10^4 \text{ s}^{-1}$ with N_2 as buffer gas and $\langle \Delta E \rangle_{\text{down}} = 390 \text{ cm}^{-1}$, the latter value corresponding to collisional efficiency $\beta_c \approx 0.25$ at 298 K. This calculated k_{diss} compares very well with the experimental values proposed by Zabel *et al.*¹ and Cox *et al.*,² namely $k_{\text{diss}} = 2.2 \times 10^4 \text{ s}^{-1}$ and $2 \times 10^4 \text{ s}^{-1}$, respectively.

Similar parameters for the isomerisation of 1-butoxy were presented in ref. 14. The RRKM value of the thermal isomerisation rate constant was calculated in this work, taking into account the tunneling effect. At 298 K, and 1 bar with threshold energy $E_0 = 43.6 \text{ kJ mol}^{-1} \equiv 3650 \text{ cm}^{-1}$, $k_{\text{iso}} = 2.1 \times 10^5 \text{ s}^{-1}$ with N_2 as buffer gas and $\langle \Delta E \rangle_{\text{down}} = 375 \text{ cm}^{-1}$ ($\beta_c \approx 0.25$ at 298 K). This value compares very well with the experimental value proposed by Cox *et al.*² namely $k_{\text{iso}} = 2.0 \times 10^5 \text{ s}^{-1}$.

Results

In the case of the 2-butoxy dissociation, the calculations of $(k_{\text{diss}})_{\text{eff}}$ from eqn (17) at the “long-time steady state”, were performed at a total pressure of 1 bar for all the experimental conditions of ref. 1 and 2, *i.e.* the different ways of producing alkoxy radicals at various temperatures and partial pressures of oxygen. The ratio $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ was then calculated as a function of the partial pressure of oxygen. These results are shown as lines in Fig. 1 for radicals produced by photolysis of the butylnitrite, and in Fig. 2 for radicals obtained from the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. In Tables 1 and 2 we compare the experimental slopes and intercepts with the calculated ones (see further below).

Similar calculations were performed at different total pressures (between 100 and 1000 mbar) and 298 K with the aim of reproducing the experiments of Zabel *et al.* (ref. 1, Fig. 3). The total pressure dependence of $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ was calculated for $P_{\text{O}_2} = 100 \text{ mbar}$ and for $P_{\text{O}_2} = P_{\text{tot}}$. The results are shown in Fig. 3 (curves) along with the experimental data (symbols). The thermal RRKM dissociation rate constant k_{diss} was calculated in the same pressure range, allowing us to obtain the calculated thermal fall-off ratio $k_{\text{diss}}/k_{\text{O}_2}$ (dashed curve).

For the purpose of a more quantitative comparison between experiment and the present calculations, both calculated and experimental results ($(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ and R_{exp} , respectively) were treated by linear least-squares, assuming linear dependence on oxygen concentration. This is shown in Tables 1 and 2, based on eqns (8) and (9) written as

$$(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2} \text{ or } R_{\text{exp}} = \text{intercept} + \text{slope} \times [\text{O}_2] \quad (19)$$

where $\text{intercept} = (k_{\text{diss}}/k_{\text{O}_2})(1 + \text{slope})$. The last two columns in the Tables 1 and 2 give $k_{\text{diss}}/k_{\text{O}_2}$ calculated from the slope and the intercept (see eqn (9)).

Very good agreement between experimental and calculated values is observed except at the higher temperatures (310 and

Table 1 2-Butoxy dissociation. Linear dependence of experimental R_{exp} and calculated ratio $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ on the partial pressure of oxygen and the resulting ratio $k_{\text{diss}}/k_{\text{O}_2}$ (see eqns (3), (8) and (9)) at several temperatures. The alkoxy radicals are obtained from photolysis of 2-butylnitrite at 360 nm. $P_{\text{tot}} = 1$ bar

T/K	$10^{-18} \times \text{Intercept/molecule cm}^{-3}$		Slope		$10^{-18} \times (k_{\text{diss}}/k_{\text{O}_2})^a/\text{molecule cm}^{-3}$	
	Experiment ^b	Calculation	Experiment ^b	Calculation	Experiment ^b	Calculation
256	0.119 ± 0.006	0.114	0.103 ± 0.007	0.098	0.109	0.108
268	0.328 ± 0.011	0.313	0.098 ± 0.005	0.109	0.299	0.282
280	0.924 ± 0.03	0.822	0.101 ± 0.006	0.122	0.839	0.733
300	3.57 ± 0.12	3.52	0.165 ± 0.024	0.147	3.06	3.07
310	6.22 ± 0.15	6.43	0.147 ± 0.079	0.162	5.42	5.6

^a See text for the difference between $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ and $k_{\text{diss}}/k_{\text{O}_2}$. ^b From Table 1 in ref. 2.

Table 2 2-Butoxy dissociation. Linear dependence of experimental R_{exp} and of the calculated ratio $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ on the partial pressure of oxygen and the resulting ratio $k_{\text{diss}}/k_{\text{O}_2}$ (see eqns (3), (8) and (9)) at several temperatures. Alkoxy radicals are obtained from reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. $P_{\text{tot}} = 1$ bar

T/K		$10^{-18} \times \text{Intercept/molecule cm}^{-3}$		Slope		$10^{-18} \times (k_{\text{diss}}/k_{\text{O}_2})^a/\text{molecule cm}^{-3}$	
		Experiment	Calculation	Experiment	Calculation	Experiment	Calculation
280 ^b		0.72	0.79	0.0635	0.037	0.677	0.76
283 ^c		1.05 ± 0.11	1.00	0.05 ± 0.02	0.048	1.00 ± 0.11	0.95
298	ref. 1	2.49	2.85	0.0843	0.060	2.29	2.67
	ref. 2	2.56 ± 0.47		0.1 ± 0.06		2.33 ± 0.45	
313 ^b		6.14	7.48	0.1121	0.099	5.52	6.82

^a See text for the difference between $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ and $k_{\text{diss}}/k_{\text{O}_2}$. ^b From Table 3 in ref. 1. ^c From Table 1 in ref. 2.

Table 3 1-Butoxy isomerisation: linear dependence of experimental $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2]$ (see text) and of the calculated ratio $(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2}$ on the partial pressure of oxygen and the resulting ratio $k_{\text{iso}}/k_{\text{O}_2}$ at several temperatures. Alkoxy radicals are obtained from photolysis of 2-butylnitrite at 360 nm. $P_{\text{tot}} = 1$ bar

T/K	$10^{-19} \times \text{Intercept/molecule cm}^{-3}$		Slope		$10^{-19} \times (k_{\text{iso}}/k_{\text{O}_2})^a/\text{molecule cm}^{-3}$	
	Experiment ^b	Calculation	Experiment ^b	Calculation	Experiment ^b	Calculation
250	0.405 ± 0.036	0.404	$(1.12 \pm 0.06) - 1$	0.12	0.362 ± 0.037	0.36
265	0.770 ± 0.046	0.743	$(1.09 \pm 0.08) - 1$	0.14	0.706 ± 0.068	0.65
278	1.40 ± 0.12	1.24	$(1.08 \pm 0.20) - 1$	0.15	1.30 ± 0.27	1.08
300	2.45 ± 0.14	2.80	$(1.15 \pm 0.24) - 1$	0.18	2.13 ± 0.46	2.37
318	4.97 ± 0.73	5.19	$(1.15 \pm 0.11) - 1$	0.20	4.32 ± 0.42	4.33

^a See text for the difference between $(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2}$ and $k_{\text{iso}}/k_{\text{O}_2}$. ^b From Table 1 in ref. 2.

313 K). One reason could be a slight underestimate of the temperature dependence of k_{O_2} ; if k_{O_2} is increased by 5%, experiments and calculation agree at 310 K but the calculated line at 313 K remains $\approx 15\%$ higher than the experimental one.

In the same way, calculations were performed for the 1-butoxy isomerisation with alkoxy radicals produced by photolysis of the corresponding nitrite. Calculated lines $(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2}$ as a function of the partial pressure of oxygen are shown in Fig. 4. As explained (*vide supra*) the experimental procedure returns the yield of butanal Y_{exp} instead of the ratio R_{exp} with the relation $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2] = (k_{\text{iso}}/k_{\text{O}_2})_{\text{eff}}$. In Table 3 we compare the experimental slopes and intercepts for $(Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2]$ vs. O_2 with the calculated ones.

$$(k_{\text{iso}})_{\text{eff}}/k_{\text{O}_2} \text{ or } (Y_{\text{exp}}^{-1} - 1) \times [\text{O}_2] \\ = \text{intercept} + \text{slope} \times [\text{O}_2] \quad (20)$$

A good agreement is observed between experimental and calculated results.

It is of interest to check the sensitivity of the calculations to the value used for the initial energy distribution of alkoxy radicals. Calculations were performed for the dissociation of 2-butoxy, at 300 K and 1 bar, for a number of delta function initial distribution between 4500 and 5000 cm^{-1} . The Table 4 compares with experiment the slopes and intercepts of the linear dependence of $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ on oxygen partial pressure. The expression calculated from the experimental² R_{exp} is equal to $(3.57 \pm 0.12) \times 10^{18} + (0.165 \pm 0.024) [\text{O}_2]$, as shown in Table 1. It is apparent that initial energy distribution in the range $5000 \pm 200 \text{ cm}^{-1}$ is still a good representation of experimental results within the experimental error. Therefore that suggests that this representation of the initial energy distribution of alkoxy radicals produced by photolysis of the corresponding nitrite at 360 nm is reasonable.

Table 4 2-Butoxy dissociation. Slopes and intercepts of the linear dependence of $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ on oxygen partial pressure for a number of delta function initial distributions IN between 4500 and 5000 cm^{-1} . $T = 300 \text{ K}$, $P = 1 \text{ bar}$

IN/cm^{-1}	4500	4800	4900	5000	Exp. ^a	5100	5200	5500
Slope	0.035	0.087	0.113	0.147	0.165	0.190	0.244	0.490
$10^{-18} \times \text{Intercept/molecule cm}^{-3}$	3.12	3.32	3.41	3.52	3.57	3.65	3.82	4.57

^a From Table 1 in ref. 2.

Discussion

The time-dependent normalised population distributions, calculated from eqn (15) at 298 K and 1 bar, are shown for chemical (Fig. 7) or photochemical (Fig. 8) production of the 2-butoxy radical. It may be observed that for times longer than about 10^{-6} s , the distribution seems thermal. However, in Fig. 9 and 10, the tail of the “long-time steady state” (time $> 10^{-4} \text{ s}$) distribution is shown on an expanded energy scale, for O_2 partial pressure equal to zero or 1 bar, along with the thermal distribution for these same temperature and total pressure. It may be noticed that regardless of the way the alkoxy radicals are produced, these final distributions are slightly higher than the thermal one, the more so the higher the O_2 partial pressure. Note the spike at 5000 cm^{-1} in Fig. 10,

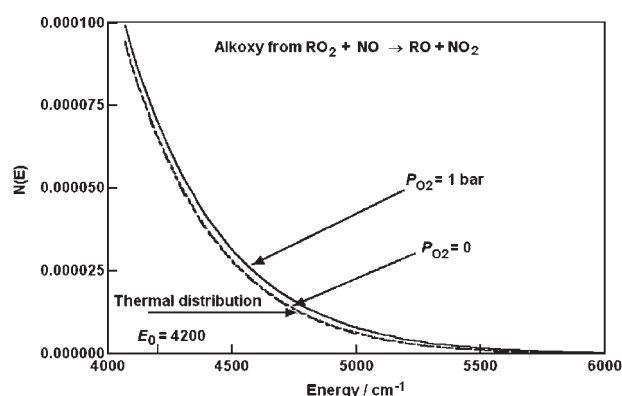


Fig. 9 Long-time steady-state population distribution $N(E)$ of alkoxy radicals produced in the reaction $\text{RO}_2 + \text{NO} \rightarrow \text{RO} + \text{NO}_2$. $T = 298 \text{ K}$; $P_{\text{tot}} = 1 \text{ bar}$. — thermal distribution; - - - $P_{\text{O}_2} = 0$; — $P_{\text{O}_2} = 1 \text{ bar}$.

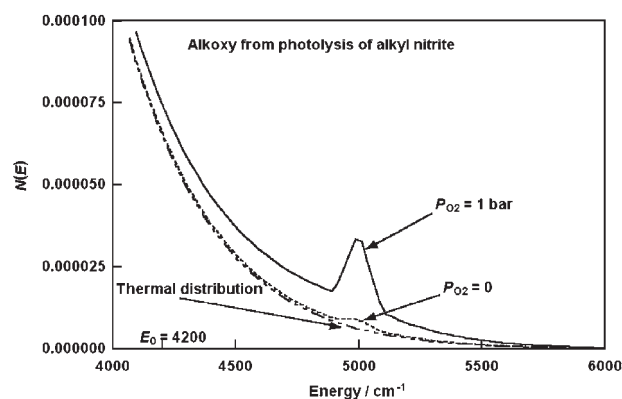


Fig. 10 Long-time steady-state distribution $N(E)$ of alkoxy radicals produced by photolysis of 2-butylnitrite at 360 nm. $T = 298 \text{ K}$; $P_{\text{tot}} = 1 \text{ bar}$. — thermal distribution; - - - $P_{\text{O}_2} = 0$; — $P_{\text{O}_2} = 1 \text{ bar}$.

the result of continuous photolysis at this energy. In consequence, the rate constant calculated from eqn (17) at this “long-time steady time” is higher than the thermal one, and increases with O_2 partial pressure, as is observed experimentally.

A very satisfactory agreement is observed between the experimental data R_{exp} and the calculated $(k_{\text{diss}})_{\text{eff}}/k_{\text{O}_2}$ for both the photochemically and chemically activated 2-butoxy radicals. Same agreement is obtained for the isomerisation of photochemically activated 1-butoxy. Therefore this work allows to validate at the same time: the theoretical model, the experimental results and the thermostructural parameters of the systems used in calculation.

Moreover, this work allows to verify the validity of our representation of the initial distribution for both the chemically and photochemically activated alkoxy radicals. This could be of a special interest for analysing, at low pressure, the experiments developed by following the temporal evolution of the alkoxy radical produced by laser photolysis of alkyl nitrite. An evaluation of the time for thermalisation of radicals could be obtained by this way.

Conclusion

In conclusion, we briefly review the successive approaches to the phenomenon studied in this work that include the concepts: hot alkoxy, prompt reaction, chemical and photochemical activation and, in our present work, “long-time steady state” in time-dependent Master Equation.

As explained in the Introduction, Peeters *et al.*⁴ and we⁵ have previously dealt with the interpretation of the experimental results of Wallington *et al.*³ by postulating the intervention of “hot alkoxy” that dissociate ‘promptly’ relative to the reaction with O_2 . This was done by calculating P_t , the ‘promptly’ dissociating fraction of chemically activated RO radicals obtained from the steady-state version of the Master Equation, which yielded what in the present context is the equivalent of $D_{\text{uni}}(t)$ at the intermediate steady-state (eqn (16) in which $k_{\text{uni}}(E)$ replaces of $k(E)$). This corresponded to the hypothesis of a stationary state between the incoming flux, on the one hand, and stabilisation and outcoming fluxes on the other. Therefore this implicitly assumed that thermal reactivation being neglected, deactivation was an irreversible step and there was a reactant sink slightly below threshold.

In 2002, Zabel *et al.*¹ observed a slight linear dependence of the measured dissociation rate constant of 2-butoxy radicals with O_2 partial pressure which they explained by the formation of chemically activated RO^* alkoxy, as mentioned in the introduction of this article. In a following publication⁶ we have developed this analysis further by showing that the remaining $(1 - P_t)$ radicals can be assigned to thermal

dissociation in competition with the reaction with O_2 . It could then be shown⁶ that the equivalent of slope in Tables 1 and 2 is “slope” = $P_t/(1 - P_t)$. In a last publication Cox *et al.*² observed the same phenomenon both with chemically and photochemically activated RO radicals.

In the present work we solve the time-dependent master equation which allows tracing out the entire time history of the same reaction sequence, from incubation to intermediate steady-state and on to the final long-time steady-state which is identified with the experimental conditions of Zabel *et al.*¹ and Cox *et al.*² At this long-time steady state, the total incoming flux equals the total outgoing flux and the net stabilisation is zero. The reservoir of thermally stabilised molecules has then grown to the point that the deactivation is precisely balanced by the thermal activation, “the well is full”. There is no temporal evolution of the energy distribution which is not the thermal distribution but a distribution displaced to higher energies since the incoming flux is at high energy. Moreover, as oxygen reacts with RO whatever the energy of this species, the low levels of energy are depleted, the more so with the higher O_2 concentration. Therefore the energy distribution is displaced to higher energies and the resulting effective dissociation or isomerisation rate constant increases with oxygen concentration.

The reader will have noticed that nowhere in our analysis of the experimental results is there any reference to the previously postulated ‘small’ amount of excited alkoxy radicals that have been assumed to be the origin of the oxygen dependence of k_{diss}/k_{O_2} . Instead we offer an alternative interpretation where (1) the oxygen dependence arises quite naturally if the transport matrix J_s takes into account that there is a “leak” of alkoxy radicals (of all energies) *via* reaction (1) or (10) with the

first-order rate constant $k_{O_2} [O_2]$, in addition to the unimolecular reaction channel which is dissociation or isomerisation, and (2) the calculation takes into account that in the cited experiments there is a continuous incoming flux of butoxy radicals.

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